

REVISION ARCHITECTURE · TRACK-1 WIN PLAN

Dukaan Saathi

Hindi-first voice + photo assistant for a kirana shop's inventory & udhaar ledger

From a working, cluster-bound prototype to a **submittable, judge-winning Hugging Face Space** — the exact, prioritized changes to win **Track 1 “Backyard AI”** of the HF × Gradio “Build Small” hackathon.

Build Small Hackathon

Track 1 — Backyard AI

Deadline 15 Jun 2026

Keep Gemma-12B → HF Space

4 killer features

8-day sprint

2026-06-07 · as-built → win plan · design only (no code changed)

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Purpose of this document. A win-focused architecture revision for the **Hugging Face x Gradio "Build Small" Hackathon — Track 1 "Backyard AI"** (deadline **2026-06-15**, ~8 days out). It explains *what the hackathon actually rewards*, where the current build stands, and the **exact, prioritized changes** that convert today's working-but-cluster-bound prototype into a *submittable, judge-winning* Hugging Face Space.

Scope: design/architecture only — **no implementation here** (per instruction). This is the blueprint we build from next. **Method:** produced from a multi-agent brainstorm (6 idea lenses → 3 scoring judges → adversarial skeptic over **52 candidates**) grounded in a code deep-dive and live hackathon research. See [Appendix B](#). **Status of inputs:** the current codebase works end-to-end (31 tests pass, live demo, 28-page as-built report). Code anchors in this doc are verified against the repo.

1. TL;DR — the win thesis

Dukaan Saathi is a Hindi-first, voice + photo assistant that runs a kirana shop's **inventory and udhaar (credit) ledger** — fully local, no cloud APIs. It is an almost-perfect fit for **Track-1 "Backyard AI"**, whose literal pitch is "*solve a real problem for... a small-business owner on your street.*"

The build already exists and works. **It is not yet a Hugging Face Space, and it has never been used by a real shopkeeper** — and those two facts are exactly what Track-1 scores hardest. So the revision is deliberately narrow:

Win thesis: Ship one **deployable HF Space** running the *unchanged* Gemma-4-12B brain, put it in the hands of **one real kirana owner with his own data**, and land **one killer feature on each of the four judging axes** — filmed as a tight Hindi demo. Everything else is cut.

Four killer features, one per axis; one deploy spine that makes the hard "live Space" rule real; a handful of small robustness fixes so nothing breaks on camera. **~50 focused hours over 8 days**, sequenced deploy-first so a valid submission exists from Day 2.

2. The hackathon, decoded

Source: <https://huggingface.co/build-small-hackathon> (researched 2026-06-07). The registration page is login-gated, so exact eligibility/team-size/licence are unconfirmed — verify on the Hub.

2.1 Format & hard rules

- **Host:** Hugging Face × Gradio. **Sponsors:** OpenBMB, OpenAI, NVIDIA, Modal, Cohere, JetBrains, Black Forest Labs.
- **Timeline:** build window **Jun 5** → **submission Jun 15, 2026**.
- **Three hard rules (non-negotiable):** 1. **Model ≤ 32B total parameters.** (*Gemma-4-12B is comfortably under.*) 2. **Must be a Gradio app hosted as a Hugging Face Space.** (*We are not on a Space yet — this is the #1 gap.*) 3. **Submission = Space link + short demo video + social-media post.**

2.2 The two tracks — we target Track 1

Track	Pitch	Our fit
1 — Backyard AI	"Solve a real problem for someone you actually know — a neighbour, a parent, a small-business owner on your street."	Bullseye. A kirana owner is the textbook user.
2 — Thousand Token Wood	Whimsical/delightful AI toys & experiences.	Not our lane.

2.3 Track-1 judging rubric (4 axes, unweighted)

#	Axis (exact wording)	What it means for us
A	"Problem is specific and real"	A named kirana owner; daily pains (challans, udhaar, stockouts) — not a generic SaaS.
B	"The person actually used it"	Real adoption is scored. Needs his <i>real data</i> in the app + evidence he used it.
C	"Honest fit between problem and the small-model constraint"	Don't look like we over-reached. Make the 12B's work <i>visibly necessary</i> (vision OCR, agentic reasoning) and document the efficiency story.
D	"Polish of the Gradio app"	A professional, fast, branded app — no spinners, no crashes on camera.

2.4 Prize & badge map (~\$48k, 29 awards)

Category	Award	Relevant to us?
Per-track	1st \$4k · 2nd \$2.5k · 3rd \$1.5k · 4th \$1k	Primary target (Track-1 podium).
Community Choice	\$2k	Helped by a great demo video + social post.
Best Agent	\$1k	Strong fit — multi-tool agentic writes/reads.

Category	Award	Relevant to us?
Tiny Titan ($\leq 4B$)	\$1.5k	<i>✗ Not pursued — we keep 12B (locked decision). Documented trade-off.</i>
Off-Brand (custom UI)	\$1.5k	<i>Decide-later — cheap CSS skin pass only.</i>
Best Demo	\$1k	The mandatory video, done well.
Sponsor: OpenBMB (MiniCPM)	\$10k	<i>✗ Would require a model swap — out of scope.</i>
Sponsor: OpenAI / NVIDIA / others	\$10k / 2× RTX 5080 / ...	<i>✗ Require their models — out of scope.</i>

Six bonus badges (more badges → Bonus-Quest Champion \$2k):

Badge	Condition	Status
Off-the-Grid	No cloud APIs	✓ Already won (fully local).
Llama-Champion	Runs via llama.cpp	✓ Already won.
Field-Notes	Wrote a report/blog	✓ Already won (28-page as-built report).
Sharing-is-Caring	Publish agent traces	<i>Decide-later</i> (trivial JSONL dump).
Off-Brand	Custom UI	<i>Decide-later</i> (CSS skin).
Well-Tuned	Publish a fine-tune	✗ Deferred ($\approx 22h$, separate project).

Headline: we already hold 3 badges and fit Best Agent **for free**. The whole game is now **rules-compliance (live Space) + the four axes**.

3. Where we stand today

3.1 As-built (works, verified)

- **Brain:** Gemma-4-12B-it **Q8_0** GGUF + mmproj, served by CUDA `llama.cpp` `llama-server` (`--jinja` , thinking-mode disabled). One model does **text + vision OCR**.
- **Agent:** deepagents 0.6.8 (LangGraph) + `ChatOpenAI` instance; **9 LangChain tools**; multi-turn memory (`InMemorySaver` + `thread_id`); Hindi system prompt → Devanagari replies.
- **Data: two SQLite DBs** — `inventory.db` (suppliers/inventory/purchases) + `transactions.db` (customers/sales/ledger); reads `ATTACH` both read-only (`inv.` / `txn.`) for cross-DB JOINS; 2-layer SELECT guard. Seeded with **synthetic** 160 SKUs / 35 customers / 120-day history.
- **Speech & vision:** faster-whisper `large-v3` (Hindi STT, no system ffmpeg); `facebook/mms-tts-hin` TTS (+ `numwords` digit→Hindi); Gemma vision OCR.
- **Proactive:** expiry watcher · udhaar reminder (LLM-drafted) · festival nudge (hardcoded 2026).
- **UI:** single-screen Gradio 6.16 (mic/photo/text + chat + live dashboard + alerts).
- **Deploy: Slurm GPU job on a cluster** — *not a HF Space*.

3.2 Honest gap audit (what would lose points)

Gap	Where	Axis at risk
Not a HF Space	no <code>Dockerfile</code> /Space config exists	Rule 2 (invalid submission)
Synthetic data only	<code>seed_*.py</code>	B (no real adoption)
Writes commit with no confirmation; oversell allowed	<code>ops.record_sale</code> → <code>new_qty<0</code> only warns (<code>ops.py:146</code>)	B, D (on-camera risk)
Ambiguous fuzzy match picks first row silently	<code>ops.find_item</code> / <code>find_customer</code>	A, B (wrong-customer udhaar)
Silent OCR failure → empty agent context	<code>normalize.ocr_image</code> returns "" (<code>normalize.py:48</code>)	D (blank reply after photo)
Double LLM call per turn (intent badge)	<code>classify_intent</code> → <code>llm.complete</code> (<code>agent.py:207</code>)	C, D (waste + latency)
Slow alerts — N sequential LLM calls	<code>proactive.run_all</code> (<code>proactive.py:232</code>)	D (sluggish demo)
Volatile chat memory; hardcoded festivals; no auth	various	minor (accept for pilot)

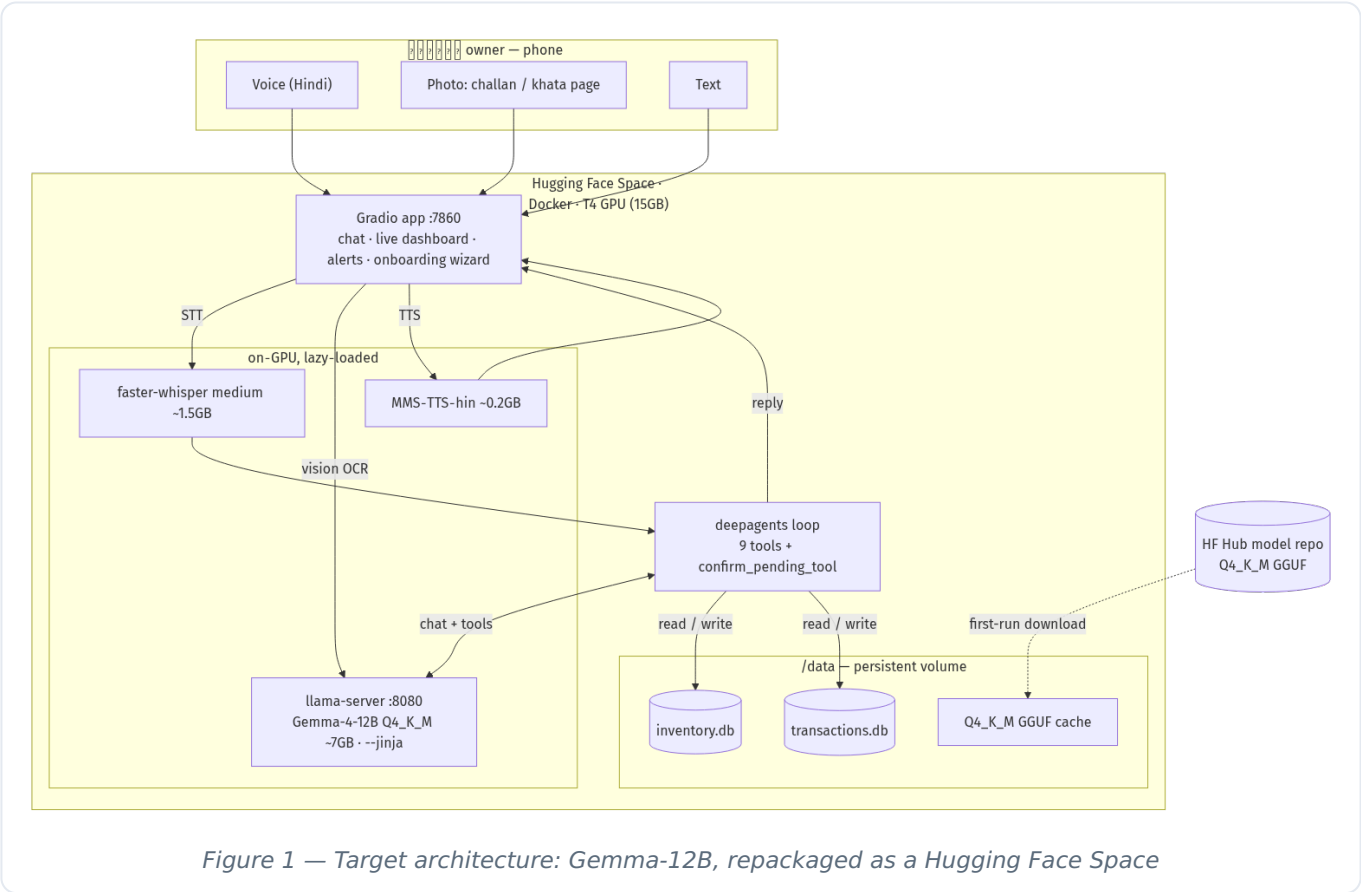
4. How Track-1 is actually scored — our move per axis

The skeptic's framing: "*Track-1 is won on the four axes, not feature count.*" So we map **one killer feature to each axis** and let the deploy spine satisfy the hard rule.

Axis	Our primary move	Backed by
A — specific & real	Challan-photo → auto-receive stock. Distributor challans are a <i>daily</i> kirana ritual; doing it by photo is unmistakably real.	Killer #1
B — actually used it	Real-owner "khata migration" onboarding → his real stock/customers in the app + a filmed pilot.	Killer #2 + \$8
C — honest small-model fit	The 12B is <i>visibly earning its size</i> : structured vision extraction from noisy Indian bills + multi-tool agentic reasoning. Plus a documented Q4 efficiency story.	Killer #1 + \$9
D — Gradio polish	One-tap morning briefing (single fast call), no cold-start spinner, confirm-before-write, branded skin.	Killer #3 + \$7

5. Revised target architecture

Nothing about the brain changes. We keep Gemma-4-12B, deepagents, the two-DB design, Whisper, and MMS-TTS. The revision **repackages** the working stack as a self-contained, GPU-backed **Docker Space**, adds a **persistent /data volume**, a **confirm-before-write** safety node, and four user-facing capabilities.



Before → after

Dimension	Today (as-built)	Revision (target)
Hosting	Slurm cluster job	HF Space (Docker, T4 GPU)
Model weights	Q8_0 (12 GB)	Q4_K_M (~7 GB) — fits T4 with headroom
STT	Whisper large-v3	Whisper medium (env-switchable) for VRAM budget
Data	synthetic seed	real owner's data via onboarding; persisted on /data
Writes	commit immediately	confirm-before-write (_PENDING + confirm_pending_tool)
Oversell	allowed (warning only)	hard-blocked with spoken Hindi refusal
Alerts/briefing	N sequential LLM calls	one LLM call → spoken "subah ka haal"
Intent badge	extra LLM call	heuristic from returned tool_calls (0 extra calls)
Cold start	n/a	progress UI + static dashboard pre-load

6. The 4 killer features

Each is scoped for the hackathon, tied to a judging axis, and has a **filmable moment**. Effort = realistic build hours.

#1 — Challan / invoice photo → auto-receive stock · 8h · Axis A + C + Best Agent

What: Owner photographs a distributor challan/bill. Gemma-4 **vision** extracts every line (product, qty, rate, supplier) as JSON; the agent loops `record_purchase_tool` per line, then **reads back** a Hindi summary. **Why it wins:** challans are a daily kirana ritual (Axis A); noisy-bill structured extraction is the 12B *visibly earning its size* (Axis C); vision → multi-tool chain in one turn is the **Best Agent** story. **Demo moment:** *snap a crumpled bill* → "राज ट्रेडर्स से 8 चीज़ें आई, कुल ₹4,320 का माल — डाल दूँ?" **Plugs into:** OCR path already exists (`normalize.ocr_image` / `describe_for_agent`); add `parse_challan()` that prompts Gemma for a JSON list, iterate `record_purchase_tool` , gate behind the confirm step (#4 must-fix). **Risk:** OCR on creased paper → mitigated by showing the extracted list before commit.

#2 — Real-owner "khata migration" onboarding · 12h · Axis B (the centrepiece)

What: A first-run wizard where the owner **speaks or photographs his existing paper bahi-khata** — items, prices, customer opening balances — and the agent ingests *his real data*, replacing the synthetic seed. Includes confirm-before-write so he trusts it from minute one. **Why it wins:** Axis B ("*the person actually used it*") hinges on **his** data being in the system; the demo video then shows **his** shop, not 160 sample SKUs (also reinforces Axis A specificity). **Demo moment:** owner reads his khata aloud → balances appear; "आपके 12 ग्राहक और 40 आइटम दर्ज हो गए।" **Plugs into:** mostly prompt engineering + a fresh `thread_id` + a new Gradio onboarding step; reuses `add_inventory` / `add_udhaar` . Pairs with the confirm node (#4).

#3 — Morning briefing (one LLM call, not N) · 4h · Best Agent + B + D

What: Replace the N-sequential-call alerts handler with **one** call: `dashboard_snapshot()` gathers all data as JSON → one Gemma prompt drafts a ~60-second spoken Hindi "**subah ka haal**" (today's stock risks, top-3 overdue customers, nearest festival). **Why it wins:** a daily habit loop (Axis B), a polished one-tap/one-playback UX (Axis D), the "intelligent agent" beat (Best Agent) — and it **kills the slow-alerts concurrency gap** for free (two-for-one). **Demo moment:** one tap → MMS-TTS speaks the day's plan in ~3s. **Plugs into:** refactor `proactive.run_all` (`proactive.py:232`) from sequential calls to a single templated completion.

#4 — Live voice udhaar round-trip · 2h · Axis A + B + Best Agent

What: Already works end-to-end — this is the **scripted hero beat**, filmed with the owner's real voice. **Demo moment:** "राजू भाई ने आज ₹350 का सामान लिया, बाद में देंगे।" → "राजू भाई का ₹350 उधार नोट हो गया — उनका कुल बाकी ₹1,150 है।" Ledger + balance update on screen instantly. **Plugs into:** zero new code; verify on the Space and use real-owner audio in the video.

Total killer-feature effort $\approx 26h$. #4 is free; #3 doubles as a robustness fix; #1 reuses the vision path; #2 is the heaviest and is the adoption centrepiece.

7. Must-fix: deployability + robustness

Deploy-first (the hard rule), then the small single-file edits that keep the write-heavy features from breaking on camera. **Total ≈ 24h.**

Deployability spine (blocks everything)

Fix	Detail	Effort
D1 • Q4_K_M requantize + CUDA Docker Space (T4)	Requantize Q8(12GB)→ Q4_K_M(~7GB) . With Whisper-medium (1.5GB) + MMS (0.2GB) it fits T4's 15GB with ~2GB headroom. CUDA <code>Dockerfile</code> (<code>nvidia/cuda:12.x-devel</code>) downloads GGUF at entrypoint → starts <code>llama-server</code> → launches Gradio (mirrors the existing sbatch flow). The foundational move.	6h
D2 • HF-Hub GGUF download + /data persistence	Host Q4_K_M on a HF model repo ; <code>hf_hub_download()</code> into the Space's <code>/data</code> volume on first start (skip on restarts). Set <code>DUKAAN_DATA_DIR=/data</code> (already supported — <code>config.py:32</code>) so both SQLite DBs persist across cold starts. <i>Pilot-critical: the owner's ledger must survive restarts.</i>	3h
D3 • Cold-start progress + static dashboard pre-load	Space cold start is 3–5 min. Poll a load-state var every 2s → Hindi progress strip; meanwhile render the dashboard from the SQLite snapshot alone (no LLM) so judges see real data at second 0 ; disable chat with a tooltip until ready.	4h

On-camera safety (protect the write features)

Fix	Detail	Effort
R1 • Confirm-before-write	New <code>confirm_pending_tool</code> + module-level <code>_PENDING[thread_id]</code> : the 5 write tools stage to <code>_PENDING</code> and return a Hindi prompt ("...₹500 उधार लिखूँ? हाँ या ना"); a <i>haan</i> commits, <i>nahi</i> discards. Highest-value trust fix — also fixes "no confirmation".	5h
R2 • Oversell hard-block	<code>record_sale</code> (<code>ops.py:143–147</code>): check <code>item.qty < qty</code> before any write; return an error + Hindi refusal (" <i>Parle-G</i> का सिर्फ़ 3 पैकेट बचा है"). ~4 lines, no schema change.	1.5h
R3 • Fuzzy-match disambiguation	<code>find_item</code> / <code>find_customer</code> : when <code>LIKE</code> returns >1 row, return <code>None</code> + candidate list so the tool asks "कौन से?". Return <code>(row_or_None, candidates)</code> ; update the 4 callers. Prevents wrong-customer udhaar.	3h
R4 • OCR-failure visibility	<code>ocr_image</code> empty (<code>normalize.py:48</code>) → <code>describe_for_agent</code> returns a Hindi "फोटो साफ़ नहीं आई — दोबारा खींचें" instead of empty context; wrap <code>vision_extract</code> with a timeout + log.	1h
R5 • Kill the double LLM call	Replace <code>classify_intent</code> (<code>agent.py:207</code>) with a heuristic over the <code>tool_calls</code> already returned by <code>run_agent</code> (write/lookup/chat). 0 extra calls, ~30% per-turn latency cut on T4.	1h

8. Real-owner pilot & adoption plan

This is the **single biggest differentiator** for Track-1 (Axis B) and the reason we asked for real-user access. A flashy app no one used loses to a plain app a real shopkeeper relies on.

Onboarding (no typing): the **khata-migration wizard** (Killer #2) ingests his real inventory + customer balances by **voice and photo**. The synthetic seed becomes a *fallback demo dataset* only.

Pilot protocol (Day 7): 1. Sit with the owner; run the wizard on **his** stock + 8-12 real customers + opening udhaar. 2. Have **him** (not us) perform 3 real tasks on his phone: record an udhaar by voice, snap a real challan, tap the morning briefing. 3. Let him run it through a real service moment if he's willing. 4. **Capture:** screen recording + over-the-shoulder video + a 20-30s **testimonial** in Hindi ("isse mera hisab aasaan ho gaya").

Adoption evidence (for judges): the testimonial clip, the pilot footage in the demo video, his real data visible in the app, and (optional, decide-later) a one-off **agent-trace export** showing his real sessions.

Demo video (<2 min, Day 8): cold-open on the real shop → onboarding in 10s → the 4 hero beats (voice udhaar, challan snap, morning briefing, an oversell refusal showing it's *safe*) → testimonial → one-line "runs fully offline on a 12B, no cloud." Then the social post + a short Field-Notes addendum.

Privacy: anonymise customer names/phones in anything published; keep his raw data on `/data` only.

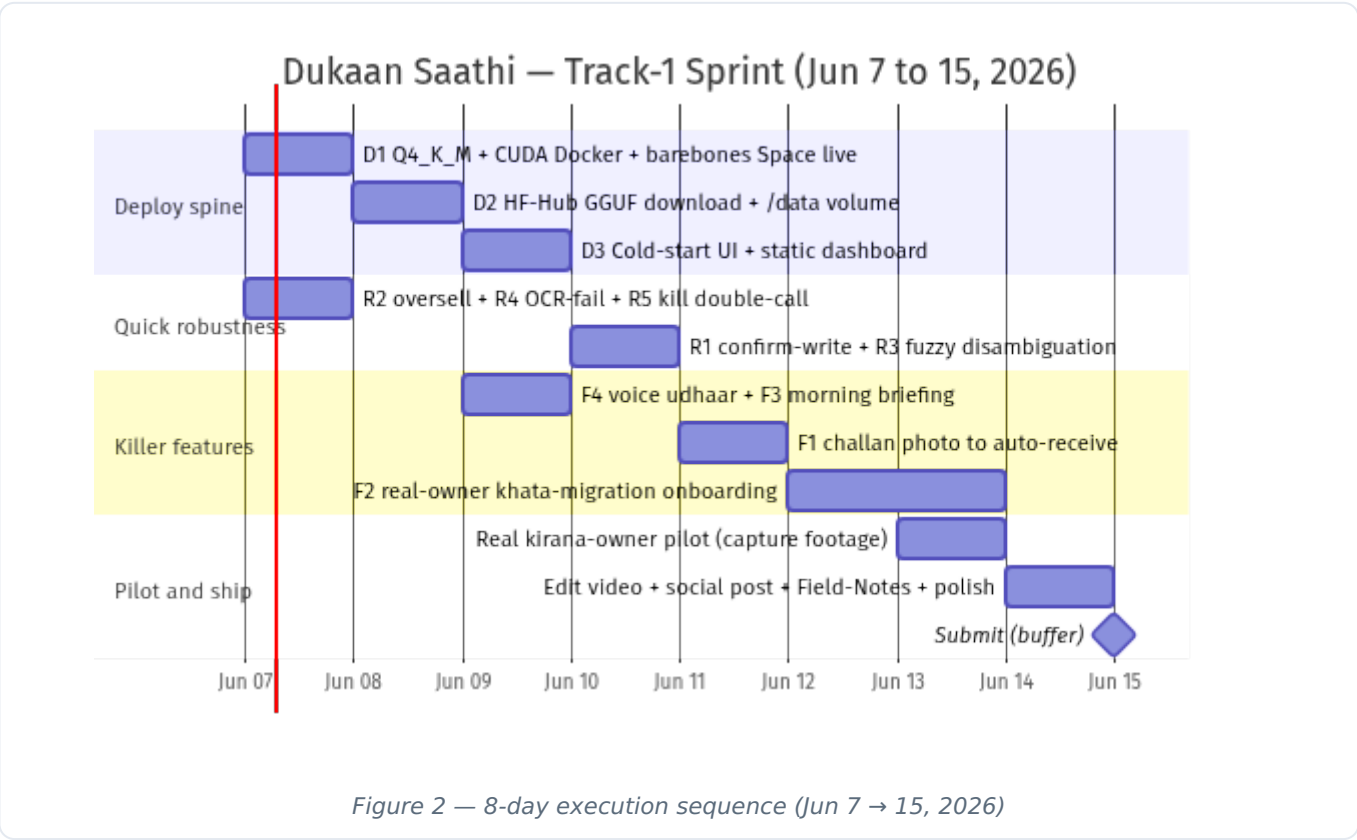
9. Deployment plan — Gemma-12B on a HF Space

Decision (locked): keep the 12B; make *it* deployable rather than shrinking the model.

- **Hardware:** HF **Space, Docker SDK, Nvidia T4 (15-16 GB)**. Apply for a **community GPU grant Day 1**; otherwise budget a paid T4 for the judging window.
- **VRAM budget:** Q4_K_M GGUF ≈ 7 GB + Whisper-medium ≈ 1.5 GB + MMS-TTS ≈ 0.2 GB $\approx \sim 9$ GB, leaving ~ 5 – 6 GB for KV-cache/headroom. (*If Q4 hurts tool-calling/Hindi, step to Q5_K_M ≈ 8.5 GB — still fits.*)
- **Serving:** keep the proven `llama-server` subprocess pattern (don't rewrite to `llama-cpp-python` / ZeroGPU — that's a 10h rearchitecture for no judge-visible gain). Entrypoint: download GGUF → start `llama-server :8080 --jinja -ngl 99` → launch Gradio `:7860`.
- **Model delivery & data:** GGUF on a HF model repo, pulled to `/data` once; DBs on `/data (DUKAAN_DATA_DIR)`, so the **owner's ledger persists** across restarts.
- **Cold start:** 3–5 min → masked by the progress UI + static dashboard (D3); models lazy-load so the page is interactive immediately.
- **Honest-fit (Axis C) artifact:** a short **Q8 vs Q4_K_M quality note** (agent battery + a few Hindi/vision cases) folded into Field-Notes — turns the quantization into a *narrative asset*, not a hidden compromise.

10. 8-day execution sequence

Sequenced **deploy-first**: a valid, submittable Space exists by **Day 2**; everything after is additive. ~6h/day.



Day	Focus	Key tasks
1 · Jun 7	Deploy spine + free wins	D1 (Q4_K_M + CUDA Dockerfile, barebones Space live with seed); apply for GPU grant; land R2/R4/R5 (3 tiny edits); validate Q4 on the agent battery.
2 · Jun 8	Make submission valid	D2 (HF-Hub GGUF download + /data persistence). A working Space now exists.
3 · Jun 9	Polish + 2 beats	D3 (cold-start UI + static dashboard); F3 (morning briefing one-call); verify F4 on the Space.
4 · Jun 10	Trust layer	R1 (confirm-before-write) + R3 (fuzzy disambiguation) — prerequisites for the write features.
5 · Jun 11	Multimodal showpiece	F1 (challan photo → auto-receive) on top of confirm + OCR-fail handling.
6 · Jun 12	Adoption centrepiece	F2 (khata-migration wizard) — build + dry-run on dummy real-style data.
7 · Jun 13	Real pilot	Finish F2; run the owner pilot; capture footage + testimonial (see §8).
8 · Jun 14	Ship	Edit <2-min video; social post; Field-Notes addendum (incl. Q4 note); optional CSS skin / trace export; final QA.

Day	Focus	Key tasks
Jun 15	Submit	Space link + video + post, with buffer.

11. Prize & badge strategy

Locked in for free: Off-the-Grid ✓ · Llama-Champion ✓ · Field-Notes ✓ · strong **Best Agent** fit ✓.

Decide-later swings (ranked by ROI — not slotted, to protect the schedule): | Swing | Effort | Payoff | Recommendation | |---|---|---|---| | **Off-Brand CSS skin** (kirana-themed Gradio theme, not a full SPA) | 1-2h | \$1.5k eligibility + lifts Axis D | **Do it** as a Day-8 polish pass. | | **Field-Notes: Q4-vs-Q8 honest-fit note** | ~1h | Strengthens Axis C narrative | **Do it** — folds into existing report. | | **Sharing-is-Caring: agent-trace JSONL export** | ~1h | Free badge + adoption evidence | **Do it** if spare hours. | | WhatsApp udhaar **send** (Twilio/WA Business) | 6h+ | Real channel, high wow | **Decide-later** — copy-to-clipboard draft already gives ~90%; external-API/sandbox risk. Only if the owner wants it. | | Voice billing / quick-sale loop | 4h | Secondary voice-write | **Defer** — voice-udhaar already proves the loop. | | **Tiny Titan / OpenBMB \$10k / Well-Tuned** | model swap / 22h | \$1.5k-\$10k | **X Not pursued** — conflict with "keep 12B" / 8-day budget. |

12. Risks & mitigations

Risk	Mitigation
T4 GPU grant not approved / cost	Apply Day 1 ; fall back to a paid T4 for the judging window only.
Q4_K_M degrades Hindi/ vision/tool-calling	Validate Day 1 on the agent battery; step up to Q5_K_M (~8.5GB, still fits) if needed.
Cold start (3-5 min) reads as broken	Progress UI + static dashboard pre-load (D3); models lazy-load.
OCR accuracy on creased challans	Confirm-before-commit shows the extracted list; owner corrects before write.
Real-owner availability slips	Lock the pilot slot now ; keep the synthetic seed as a fallback demo dataset; record asynchronously if needed.
Scope/time overrun	Deploy-first ⇒ valid submission from Day 2; features are additive; onboarding can shrink to photo-import-only if time.
deepagents recursion/ subagent bug	Already mitigated (no subagents, <code>recursion_limit=60</code>) — do not add subagents.
Wrong-customer udhaar / oversell on camera	R1/R2/R3 land before the pilot/film day.

13. Explicitly deferred

Cut to protect the 8-day plan (good ideas, wrong week). Grouped from the 41 pruned candidates: - **Second/third vision flows:** UPI-screenshot reconciliation, shelf-stockout audit, product-label/ expiry scan — *avoid multimodal sprawl; one vision wow (challan) is enough.* - **External integrations:** WhatsApp/Twilio send, ONDC order management, UPI soundbox webhook — *external-API risk.* - **Analytics niceties:** credit-risk badge, slow-mover discounter, peak-hours, churn/retention, competitor price check. - **Supplier:** voice reorder "same as last time", distributor-visit prep, QR udhaar card. - **Heavier infra:** cross-DB atomic writes (WAL) (*2h compensating-rollback may ride along with R2 if time*), persistent chat memory, async sentence-streaming TTS, `llama-cpp-python` /ZeroGPU rearchitecture. - **Big bets:** Well-Tuned Whisper fine-tune ($\approx 22h$), full Off-Brand React/Vue SPA ($\approx 14h$), model swap for sponsor prizes.

14. Appendices

Appendix A — Locked decisions (this revision's guardrails)

- 1. **Keep Gemma-4-12B** as the primary model; make it Space-deployable (Q4_K_M). 4B only as a documented fallback.
- 2. **Real kirana-owner pilot** is the adoption centrepiece (real data, no typing).
- 3. **Win-focused scope only** — 4 killers + must-fix; everything else deferred.
- 4. **Prize swings = decide-later** (Off-Brand skin / traces / Field-Notes note ranked above).

Appendix B — How this blueprint was produced

A background multi-agent workflow: **6 idea lenses** (judging-axis maximizer · killer demo moments · real-pilot enabler · 12B-on-a-Space deployability · must-fix robustness · prize-swing strategist) → **52 candidates** → a **3-judge scoring panel** (rubric / builder / showman) → an **adversarial skeptic** that pruned to **4 keep + 8 must-fix + 41 cut**. Grounded in a code deep-dive + live hackathon research. *(The final formatting agent was interrupted; results were recovered from the workflow journal and synthesized here.)*

Top candidates by aggregate panel score (axis / feasibility / demo / prize, out of 5): |
Candidate | Axis | Feas | Demo | Prize | |---|---|---|---|---| | Real-data onboarding (owner speaks his inventory) | 5.0 | 5.0 | 5.0 | 3.7 | | Challan snap → auto-receive | 5.0 | 4.0 | 5.0 | 4.7 | | Morning briefing (auto-summary) | 4.0 | 5.0 | 5.0 | 4.0 | | Live voice udhaar round-trip | 4.3 | 5.0 | 5.0 | 3.7 | | Voice confirm-before-write | 4.3 | 3.7 | — | — |

Appendix C — Verified code anchors (where the work lands)

`config.py:32 DATA_DIR / DUKAAN_DATA_DIR` · `config.py:34-35` DB paths · `config.py:65 WHISPER_MODEL=large-v3` · `ops.py:129 record_sale` (oversell at :146) · `ops.py find_item / find_customer` · `normalize.py:48` OCR empty-return · `agent.py:207 classify_intent` extra call · `agent.py:171 tool_calls` already collected · `proactive.py:232 run_all` sequential. No `Dockerfile` / Space config exists yet.

End of revision. No code has been changed. Next step on approval: turn \$6/\$7/\$10 into an implementation plan.